Gift A. Nwatuzie

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LinkedIn | GitHub | Portfolio

Professional Summary

Software Engineer with 8+ years of experience designing, developing, and maintaining scalable backend systems, distributed software, and AI-powered applications. Strong expertise in Python, Java, C++, RESTful API development, software architecture, distributed machine learning, and intelligent systems. Experienced building production-oriented software, applying machine learning to real-world engineering problems, and collaborating with software engineers, applied researchers, and cross-functional teams to deliver reliable, maintainable solutions. Currently pursuing a Ph.D. in Computer Science with research focused on trustworthy AI, distributed machine learning, and secure intelligent systems, expected December 2026.

Technical Skills

Programming	Python, Java, C++, SQL, JavaScript, HTML/CSS
Software Engineering	Backend Development, Distributed Systems, REST APIs, Software Architecture, System Design, Object-Oriented Programming, Docker, Git, CI/CD, Linux
Machine Learning	PyTorch, TensorFlow, Scikit-learn, Deep Learning, Federated Learning, Explainable AI, Model Evaluation, Responsible AI
Data Engineering	Pandas, NumPy, Feature Engineering, Statistical Analysis, Experimental Design, Data Visualization
Security & Distributed AI	Differential Privacy, Secure Aggregation, Homomorphic Encryption, Distributed Machine Learning, Edge AI

Software Engineer
May 2019 – July 2026

Tech Nova Solutions — *Remote*

- Designed, developed, and maintained scalable backend software and distributed services supporting data-intensive applications and intelligent systems.
- Engineered secure RESTful APIs and backend components emphasizing reliability, maintainability, and production readiness.
- Collaborated with software engineers, project managers, and stakeholders to translate business requirements into scalable software solutions.
- Automated software delivery using Git-based workflows, testing frameworks, and CI/CD pipelines to improve deployment consistency and software quality.
- Applied machine learning techniques to software engineering and cybersecurity challenges while integrating explainability, privacy, and security considerations into production-oriented workflows.

Software Engineering Fellow
Mar 2021 – Sep 2022

Coding Expert Bootcamp — *Remote*

- Designed and implemented a recommendation-system prototype in Python using machine learning, explainability techniques, and fairness constraints.
- Built machine learning pipelines to evaluate recommendation quality, fairness, and model transparency.
- Collaborated with multidisciplinary teams to translate research concepts into practical software solutions.
- Mentored junior software engineers on software architecture, machine learning engineering, and responsible AI development.

Software Engineer
Jan 2020 – Feb 2021

Trendsville / Pack Media Ltd — *Remote*

- Designed and implemented backend software components supporting web applications and internal business services.
- Developed secure RESTful APIs and optimized backend functionality to improve application reliability and maintainability.
- Collaborated with cross-functional teams to translate business requirements into scalable software solutions.
- Applied DevOps practices, automated testing, version control, and continuous integration principles to improve software quality.
- Integrated machine learning explainability and bias-detection techniques into analytics workflows to improve transparency of intelligent software systems.

Software Engineer
Jan 2016 – Dec 2019

Civil Service Commission — *Nigeria*

- Designed, developed, and maintained software applications supporting public-sector operational workflows.
- Automated manual business processes, improving efficiency, system reliability, and data consistency across multiple departments.
- Implemented secure software development practices while improving application performance and maintainability.
- Collaborated with technical teams to troubleshoot software issues, deploy enhancements, and improve overall system stability.
- Mentored junior developers on software engineering best practices, debugging techniques, and code quality.

Adjunct Instructor
Jan 2026 – Present

Kent State University — *Kent, OH*

- Teach undergraduate and graduate courses in algorithms, software engineering, and computational problem solving using Python, Java, and C++.
- Mentor students in software design, debugging, algorithm optimization, and engineering best practices.

Graduate Teaching Assistant
Aug 2022 – Dec 2025

Kent State University — *Kent, OH*

- Supported programming laboratories, debugging sessions, and software engineering instruction while mentoring students in algorithm implementation and software development.

Graduate Hourly Employee Coordinator
Jan 2024 – Dec 2025

Kent State University — *Kent, OH*

- Coordinated technical operations, project priorities, and communication among faculty, staff, and student employees while supporting departmental initiatives.

Selected Machine Learning & Software Engineering Projects

Federated Learning for Secure Healthcare Analytics — *Distributed Machine Learning*

- Designed and implemented a distributed federated-learning framework integrating differential privacy, secure aggregation, encrypted model updates, and communication-efficient training.
- Developed scalable machine learning pipelines for privacy-preserving healthcare prediction across decentralized data sources.
- Achieved 93.5% prediction accuracy on the Framingham dataset and 92.1% on the Cleveland Heart Disease dataset, with five-fold cross-validation averaging 92.8% $\pm$ 1.3%.
- Evaluated robustness using membership-inference and gradient-reconstruction attack simulations while reducing communication overhead through model-update sparsification.

Bias-Aware Recommendation System Prototype — *Machine Learning / Recommendation Systems*

- Designed and implemented a recommendation-system prototype using Python, machine learning, explainability techniques, and fairness constraints.
- Developed end-to-end machine learning pipelines supporting recommendation generation, model evaluation, and explainable decision making.
- Evaluated recommendation quality using fairness metrics and explanation techniques to improve transparency and responsible personalization.
- Collaborated with multidisciplinary teams to transform research concepts into practical software engineering solutions.

Machine Vision and IoT Sensing Framework — *Computer Vision / Edge AI*

- Designed and developed a multimodal software framework integrating computer vision models with IoT sensor data for real-time intelligent monitoring.
- Built deep learning models for image classification and segmentation achieving 94.8% disease-classification accuracy and 87.6% mean IoU for weed segmentation.
- Evaluated deployment performance on edge-computing hardware while optimizing inference speed and resource utilization.
- Integrated multimodal sensor information to improve prediction reliability across heterogeneous environments.

Deep Learning for Network Monitoring — *Cybersecurity / Intelligent Systems*

- Designed deep learning models for anomaly detection and intelligent data recovery within network-monitoring environments.
- Developed predictive models capable of identifying abnormal network behavior while improving recovery accuracy and operational reliability.

- Evaluated system performance using benchmark cybersecurity datasets and multiple machine learning evaluation metrics.
- Applied scalable deep learning techniques to improve network resilience and intelligent monitoring capabilities.

FairAudit: Causal Analysis for Intersectional Bias *Responsible AI*

- Developed a causal machine learning framework for diagnosing and mitigating intersectional bias using Shapley-based causal attribution.
- Designed subgroup-aware mitigation strategies that reduced intersectional disparity by 52% while maintaining predictive performance.
- Evaluated the framework using ACSIncome and CelebA datasets with fairness, accuracy, and subgroup-disparity metrics.
- Improved transparency of machine learning models through explainable AI techniques supporting responsible AI deployment.

Technical Leadership

Lead Software Architect & Ethical AI Advisor — Jan 2021 – Dec 2022

Techvision Volunteer — *Remote*

- Led the architecture and development of a mobile-first healthcare platform supporting secure digital health services for underserved communities.
- Designed offline-first software workflows, secure cloud synchronization, automated validation, and real-time consistency checks, improving reported rural healthcare data reliability by approximately 70%.
- Collaborated with multidisciplinary teams to deliver scalable software solutions while coordinating technical implementation across contributors.
- Co-authored Responsible AI guidelines promoting fairness, transparency, privacy, and ethical software development.
- Mentored software developers on software architecture, secure system design, engineering best practices, and responsible AI implementation.

Education

Ph.D. in Computer Science — Expected December 2026
Kent State University, Kent, Ohio
Relevant Coursework: Machine Learning, Artificial Intelligence, Distributed Systems, Advanced Algorithms, Computer Security

M.Sc. in Computer Systems Engineering — 2019
University of East London

B.Sc. in Software Engineering — 2018
University of East London

Selected Publications

- Nwatuzie, G., & Peyravi, H.
  "A Framework for Integration of Machine Vision with IoT Sensing."
  Sensors, 2025.
- Nwatuzie, G., & Peyravi, H.
  "DRNMS: An Enhanced Deep Learning-Based System for Data Recovery and Anomaly Detection in Network Monitoring."
  Springer, 2024.
- Nwatuzie, G., & Peyravi, H.
  "Quantum-Enhanced Intrusion Detection: A Novel Hybrid Approach Using Quantum Deep Learning."
  Springer, 2025.

Awards

- NSF Travel Award (2025)
- Award for Excellence in Innovation (2020)
- STEM Outstanding Achievement Award (2018)